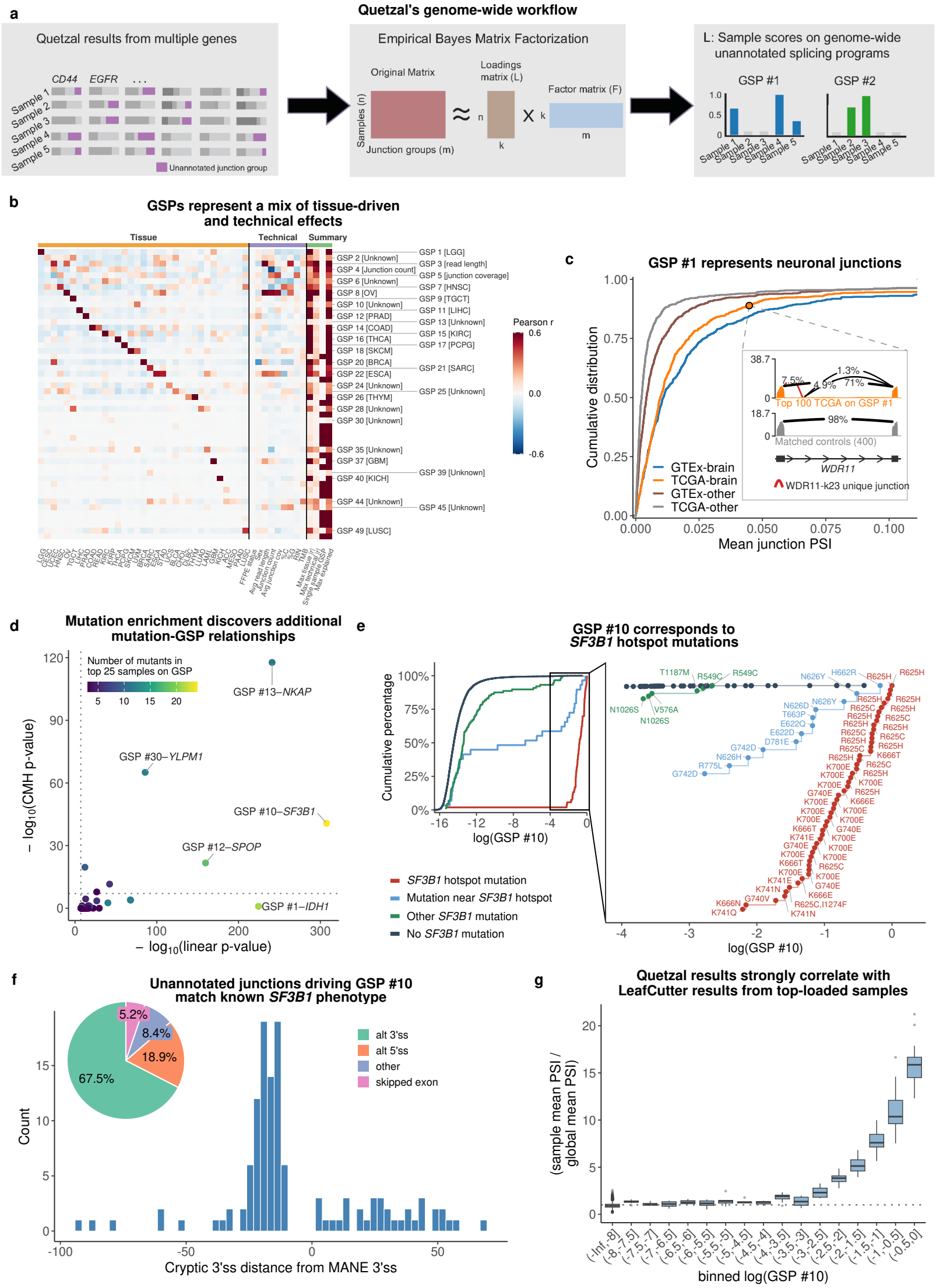

a
Quetzal's genome-wide workflow
Quetzal results from multiple genes
CD44
EGFR
Sample 1
Sample 2
Sample 3
Sample 4
Sample 5
Unannotated junction group
Empirical Bayes Matrix Factorization
Original Matrix
Loadings matrix (L)
Factor matrix (F)
Samples (n)
Junction groups (m)
L: Sample scores on genome-wide unannotated splicing programs
GSP #1
GSP #2
b
GSPs represent a mix of tissue-driven and technical effects
Tissue
Technical
Summary
GSP 1 [LGG]
GSP 2 [Unknown]
GSP 3 [read length]
GSP 4 [Junction count]
GSP 5 [junction coverage]
GSP 6 [Unknown]
GSP 7 [HNSC]
GSP 8 [OV]
GSP 9 [TGCT]
GSP 10 [Unknown]
GSP 11 [LIHC]
GSP 12 [PRAD]
GSP 13 [Unknown]
GSP 14 [COAD]
GSP 15 [KIRC]
GSP 16 [THCA]
GSP 17 [PCPG]
GSP 18 [SKCM]
GSP 20 [BRCA]
GSP 21 [SARC]
GSP 22 [ESCA]
GSP 24 [Unknown]
GSP 25 [Unknown]
GSP 26 [THYM]
GSP 28 [Unknown]
GSP 30 [Unknown]
GSP 35 [Unknown]
GSP 37 [GBM]
GSP 39 [Unknown]
GSP 40 [KICH]
GSP 44 [Unknown]
GSP 45 [Unknown]
GSP 49 [LUSC]
Pearson r
0.6
0.0
-0.6
c
GSP #1 represents neuronal junctions
Cumulative distribution
Mean junction PSI
GTEx-brain
TCGA-brain
GTEx-other
TCGA-other
Top 100 TCGA on GSP #1
Matched controls (400)
WDR11
WDR11-k23 unique junction
d
Mutation enrichment discovers additional mutation-GSP relationships
Number of mutants in top 25 samples on GSP
GSP #13–NKAP
GSP #30–YLPM1
GSP #10–SF3B1
GSP #12–SPOP
GSP #1–IDH1
– log10(CMH p-value)
– log10(linear p-value)
e
GSP #10 corresponds to SF3B1 hotspot mutations
Cumulative percentage
log(GSP #10)
SF3B1 hotspot mutation
Mutation near SF3B1 hotspot
Other SF3B1 mutation
No SF3B1 mutation
f
Unannotated junctions driving GSP #10 match known SF3B1 phenotype
5.2%
8.4%
18.9%
67.5%
alt 3'ss
alt 5'ss
other
skipped exon
Count
Cryptic 3'ss distance from MANE 3'ss
g
Quetzal results strongly correlate with LeafCutter results from top-loaded samples
(sample mean PSI / global mean PSI)
binned log(GSP #10)